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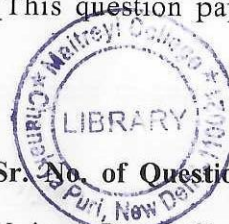
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Section III

1. Find the mass of a wire in the shape of curve $C: x = 3 \sin t, y = 3 \cos t, z = 2t$ for $0 \leq t \leq \pi$ and density at point (x, y, z) on the curve is $\delta(x, y, z) = z$.
2. Find the work done by force $\vec{F} = x\hat{i} + y\hat{j} + (xz - y)\hat{k}$ on an object moving along the curve C given by $R(t) = t^2\hat{i} + 2t\hat{j} + 4t^3\hat{k}$.
3. Use Green's theorem to find the work done by the force field $\vec{F}(x, y) = y^2\hat{i} + x^2\hat{j}$ when an object moves once counterclockwise around the circular path $x^2 + y^2 = 2$.
4. State and prove Green's Theorem.
5. Evaluate $\oint (2xy^2z \, dx + 2x^2yz \, dy + (x^2y^2 - 2z) \, dz)$ where C is the curve given by $x = \cos t, y = \sin t, z = \sin t, 0 \leq t \leq 2\pi$ traversed in the direction of increasing t .
6. Use divergence theorem to evaluate $\iint_S \vec{F} \cdot \vec{N} \, ds$ where $\vec{F} = (x^5 + 10xy^2z^2)\hat{i} + (y^5 + 10yx^2z^2)\hat{j} + (z^5 + 10zy^2x^2)\hat{k}$ and S is closed hemisphere surface $z = \sqrt{1 - x^2 - y^2}$ together with the disk $x^2 + y^2 \leq 1$ in x - y plane.

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[This question paper contains 4 printed pages.]



04.01.2024(M)
Your Roll No.....

Sr. No. of Question Paper : 4518

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Unique Paper Code : 32351303

Name of the Paper : Multivariate Calculus

Name of the Course : B.Sc. (H) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All sections are compulsory.
3. Attempt any Five questions from each section.
4. All questions carry equal marks.

Section I

1. Find the following limits :

(i) $\lim_{(x,y) \rightarrow (0,0)} (1 + x^2 + y^2)^{\frac{1}{x^2 + y^2}}$

(ii) $\lim_{(x,y) \rightarrow (0,0)} x \log \sqrt{x^2 + y^2}$

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2. Find an equation for each horizontal tangent-plane to the surface

$$z = 5 - x^2 - y^2 + 4y$$

3. The output at a certain factory is $Q = 150K^{\frac{2}{3}}L^{\frac{1}{3}}$ where K is the capital investment in units of \$1000, and L is the size of Labor force measured in worker-hours. The current capital investment is \$500,000 and 150 worker hours of Labor are used. Estimate the change in output that results when capital investment is increased by \$500 and Labor is decreased by 4 worker-hours.

4. Let $w = f(t)$ be a differentiable function of t where $t = (x^2 + y^2 + z^2)^{1/2}$. Show that

$$(dw/dt)^2 = (\partial w/\partial x)^2 + (\partial w/\partial y)^2 + (\partial w/\partial z)^2.$$

5. Let $f(x, y, z) = xyz$ and let $\hat{u}$ be a unit vector perpendicular to both $\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} - \hat{k}$. Find the directional derivative of f at $P_0(1, -1, 2)$ in the direction of $\hat{u}$.

6. Find the absolute extrema of the function $f(x, y) = e^{x^2 - y^2}$ over the disk $x^2 + y^2 \leq 1$.

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Section II

1. Evaluate the double integral $\iint_D \frac{dA}{y^2 + 1}$ where D is triangle bounded by $x=2y$, $y=-x$ and $y=2$.

2. Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} y \sqrt{x^2 + y^2} dy dx$ by converting to polar coordinates.

3. Find the volume of tetrahedron T bounded by plane $2x + y + 3z = 6$ and co-ordinate planes.

4. Use spherical co-ordinates to verify that volume of a half sphere of radius R is $\frac{2}{3}\pi R^3$.

5. Use cylindrical co-ordinates to compute the integral $\iiint_D z(x^2 + y^2)^{-\frac{1}{2}} dx dy dz$ where D is the solid bounded above by the plane $z=2$ and below by the surface $2z = x^2 + y^2$.

6. Use a suitable change of variables to compute the double integral $\iint_D \left(\frac{x-y}{x+y}\right)^2 dy dx$, where D is the triangular region bounded by line $x + y = 1$ and co-ordinate axes.